

Reproducing the LHS 1140 b helium line with a solved wind: how far EXHALE gets, and what is missing

1. Goal, and the answer up front

The objective is to reproduce the measured He I 10830 Å transit line of LHS 1140 b with EXHALE, which solves the wind rather than imposing it, and to establish what — if anything — has to be added to the physics for the model to match the observation.

Result. Two statements carry the memo, and they point in opposite directions: the *amount* of absorption can be matched under every assumption combination tried, but only at a composition that those assumptions move by a factor 80; the *width* cannot be matched under any of them, and the velocity it asks for is the same number every time.

(i) **The amount is always matchable, so the composition it implies is a conditional statement rather than a measurement.** Every combination of assumed spectrum and assumed mixing reaches the observed red-pair equivalent width at some helium abundance, and the escape rate — an output, not a fit — barely moves along the way: $\log_{10}(\dot{M}/\text{g s}^{-1}) \simeq 7.6\text{--}7.8$ on the GJ 1132 proxy spectrum across composition, diffusion and eddy coefficient alike, and 8.6 on the GJ 699 proxy, the second stand-in for LHS 1140's unobserved XUV, 5.1 times brighter, which the paper also uses — there within 0.04 dex of what the paper itself retrieves on that same proxy (Sect. 10.). What the assumptions move is where the crossing lands. In the well-mixed limit it is He/H = 0.55 on the GJ 1132 proxy and 0.060, below solar, on the GJ 699 one. With binary H/He element diffusion active — helium then settles out of the base unless an eddy term lifts it — it rises again, to 4.68 wherever $K_{zz} \leq 10^5 \text{ cm}^2 \text{ s}^{-1}$ does nothing against the molecular coefficient, 2.09 at the adopted $K_{zz} = 10^9$ and 1.11 at 10^{11} (Sect. 8.3). End to end that is 0.060 to 4.68, a factor 80, over choices this measurement cannot arbitrate: strongly conditional on the assumed spectrum, weakly and monotonically on the assumed mixing, as $K_{zz}^{-0.14}$ wherever the eddy term acts at all. Any composition read off this line — the paper's H:He $\lesssim 10^{-3}$ as much as any value here — inherits both conditions.

(ii) **The width is not reached on any axis, and the deficit is invariant.** At the equivalent-width crossing the profile is about 3.4 times too deep and 3.4 times too narrow, and no composition fixes it, because composition changes how much helium absorbs, not how the absorption is spread in velocity. Measured directly — convolving the solved profile with a Gaussian and asking which one reproduces the observed line — the missing spread is 22.30 km s^{-1} FWHM ($\sigma = 9.5 \text{ km s}^{-1}$) in the well-mixed limit, 22.31 km s^{-1} with diffusion and the adopted K_{zz} , and 22.08 km s^{-1} on the $5.1 \times$ brighter spectrum (Sects. 8.2, 8.3 and 10.). Supplied as one Gaussian it puts depth, blue depth and red/blue ratio inside the observed 1σ interval at once, with nothing tuned to them; what produces it is unidentified. It is not thermal — the solved gas cools below 1000 K over the line-forming region — and not bulk outflow, which reaches 1.2 km s^{-1} at $20R_p$ on the fainter spectrum and 3.6 on the brighter. Read as turbulence it is $\mathcal{M} = 2.4\text{--}4.2$ against the sound speed these solutions themselves carry, supersonic on every convention and well above what the literature is willing to assume. The one mechanism that would supply width without a free parameter — the adiabatic-cooling metastable bump of Schulik & Owen (2025) — is present in these solutions and measured, but sits at $0.3\text{--}1.4 \text{ km s}^{-1}$ (Sect. 8.5). The shortfall is in the energy and momentum budget, not in the composition, and it is a published problem rather than a defect of this

setup (Sect. 8.).

The paper’s own p-winds fit does not reproduce the line either: it carries roughly the right width, because its temperature is imposed at 5000–6000 K where the line forms, but overshoots the blue component threefold. Adding broadening does not rescue it: its absorbing power is 24% below the measurement, so no convolution can reach the observed depth and width at once.

Everything below uses inputs verified against the authors’ own released material (Zenodo 20723095): their p-winds driver script, their Mega-MUSCLES v23 spectra, their reduced transmission spectrum and their likelihood grid. Products live in LHS1140b/; the notebook LHS1140b/LHS1140b_analysis.ipynb regenerates every number and figure here.

2. Inputs, and how each was pinned

2.1 System

Planet and star from Cadieux et al. (2024) as quoted by C26: $R_p = 1.730 \pm 0.025 R_\oplus$, $M_p = 5.60 \pm 0.19 M_\oplus$, $T_{\text{eq}} = 226$ K, $a = 0.0946$ AU, $M_\star = 0.1844 M_\odot$, $R_\star = 0.2159 R_\odot$. One trap: EXHALE converts R_p with its own internal constants (mean radii, $R_J = 6.9911 \times 10^9$ cm, $R_\oplus = 6.3725 \times 10^8$ cm), so the input must be built with those and not with IAU equatorial values — a 2.2% error in radius, 4.4% in transit depth, if taken from the wrong pair.

2.2 Lower boundary condition

EXHALE anchors the lower boundary at the planetary radius R_p using the standard CETIMB pressure-anchored condition (Base BC: pressure 1.0), fixing the base pressure at $P_0 = 1.0 \mu\text{bar} = 1.0 \text{ dyn cm}^{-2} = 1.0 \text{ erg cm}^{-3}$ (0.1 Pa). The corresponding base total number density n_0 follows from the ideal gas law at the equilibrium temperature $T_0 = T_{\text{eq}} = 226.0$ K:

$$n_0 = \frac{P_0}{k_B T_0 n_{\text{tot},\text{bc}}} = \frac{1.0 \text{ dyn cm}^{-2}}{(1.380649 \times 10^{-16} \text{ erg K}^{-1}) \times 226.0 \text{ K}} = 3.20485 \times 10^{13} \text{ cm}^{-3}, \quad (1)$$

where $n_{\text{tot},\text{bc}} = 1.0$ for neutral atomic gas. In logarithmic form, $\log_{10}(n_0/\text{cm}^{-3}) = 13.506$, which is entered in `input.inp` as `Log10 lower boundary number density [cm^-3]: 13.506`. When Base BC: pressure is specified, EXHALE dynamically re-evaluates and enforces this relation internally (`input_read.f90`).

2.3 Stellar spectrum

The paper describes its normalization two ways that cannot both hold: the Supplement’s prose says the proxy SED is scaled so its integrated X-ray flux matches that measured for LHS 1140, while the quoted constants ($A_{1132} = 0.59$, $A_{699} = 0.056$) are the catalog X-ray flux *ratios*, a factor 1.8 apart. The authors’ script settles it:

```
Lx_ratio_1132 = 0.59
```

```
norm_constant_1132 = Lx_ratio_1132*(d_1140/(smax_1140*au2m))*2
```

applied to the plain `const-res v23` product. That combination also reproduces the fiducial XUV flux the main text quotes, 0.033 W m^{-2} (we measure $33.6 \text{ erg s}^{-1} \text{ cm}^{-2}$ over 10–1300 Å). All results here use it. Note the proxy SED and

the paper’s own APEC fit disagree in the X-ray band (4.4 against $2.9 \text{ erg s}^{-1} \text{ cm}^{-2}$ at the orbit); that inconsistency is inherited from the paper’s procedure, not introduced here.

2.4 Measurement

The comparison target is the authors’ released Fig. 3B spectrum (129 samples, vacuum, planetary rest frame, correlated-noise corrected), not a reading of a figure. Fitting it with our own three-Gaussian extractor (`he_line_metrics.py`, the same one applied to every model spectrum here) reproduces the paper’s MCMC to within its published uncertainties:

	this extractor	paper
blended red depth	1.254%	$1.24^{+0.22}_{-0.23}\%$
blue depth	0.198%	$0.25^{+0.14}_{-0.12}\%$
red/blue	6.35	$6.7^{+12.7}_{-3.1}$
FWHM	$0.841 \text{ \AA}$	$0.86^{+0.15}_{-0.27} \text{ \AA}$

so models and measurement can be compared through one instrument.

3. Reproducing the paper’s p-winds model

Run at the authors’ settings — their SED normalization, $R_p = 0.154 R_J$, $b = 0.23$, `turbulence_broadening=True`, five-phase transit averaging, and their fitted line-of-sight bulk velocity $v_{\text{wind}} = 2.26 \text{ km s}^{-1}$ — and at the MCMC medians ($\dot{M} = 2.03 \times 10^8 \text{ g s}^{-1}$, $T = 5160 \text{ K}$, $\text{H}:\text{He} = 1.01 \times 10^{-3}$), the vector their Fig. 4 legend states for the plotted curve, our reproduction gives a red depth of 1.376% and FWHM $0.575 \text{ \AA}$ against the 1.32% and $\sim 0.71 \text{ \AA}$ read off their Fig. 4: a 3% match in depth, but a line about 20% too narrow.

The width residual is not a consequence of which point in parameter space is drawn. Running the same reproduction at the grid maximum-likelihood point ($1.83 \times 10^8 \text{ g s}^{-1}$, 5131.6 K , 8.338×10^{-4}) gives red 1.366% and FWHM $0.581 \text{ \AA}$, i.e. the two vectors differ by less than $0.01 \text{ \AA}$ in width. What produces the extra width in the published curve has not been identified.

Two points are worth recording because both cost time to find.

First, the curves drawn over the spectrum in the paper’s Fig. 3 are the *three-Gaussian empirical fits*, not p-winds; the physical model is overlaid only in Fig. 4. Fig. 3’s apparently perfect agreement is a fit to the data, and says nothing about the model’s line shape.

Second, the paper’s plotted model is visibly broader than a plain p-winds run, which we initially modelled as an undocumented instrumental kernel. It is not: their script sets `turbulence_broadening=True` and averages over five transit phases. Both widen the line, and together they account for the difference.

4. Turbulent broadening: what the literature does

Since that switch turned out to matter, all 43 papers under references/ were checked: every PDF converted to text, searched for `turbulen`, `microturbulen` and `non-thermal broadening`, and each hit read in context. Fifteen matched;

most use “turbulent” for eddy diffusion K_{zz} or for flow structure in 3D simulations, not for line broadening. Section 4.1 tabulates the ones that bear on broadening.

The convention, stated most explicitly by Yan et al. (2024), replaces the thermal Doppler width by

$$\Delta\nu_D = \frac{\nu_0}{c} \left(\frac{2kT}{m} + v_{\text{turb}}^2 \right)^{1/2}, \quad v_{\text{turb}} = \sqrt{\frac{5kT}{3m}} = c_s, \quad (2)$$

the monatomic sound speed, as assumed by Salz et al. (2018), Lampón et al. (2020) and Czesla et al. (2022). The p-winds implementation uses $\sqrt{5kT/6m}$ in a σ -convention, which is the same physics expressed against a Doppler width that lacks the factor 2; in EXHALE’s b -parameter convention the same term is $b \rightarrow \sqrt{11/6} v_{\text{th}}$, a factor 1.354. That is what EXHALE_TRANSIT_TURB=1 applies. It is opt-in and off by default, because the field does not agree that it belongs.

4.1 Paper by paper

Adopt it as an assumed value. Salz et al. (2016), Eq. (5): “ b is the Doppler parameter, v_{th} the thermal Doppler velocity, and for the microturbulence v_{micro} we use the sound speed”, applied when synthesizing the transit profile. (Their atmosphere solver separately carries a constant 1 km s^{-1} microturbulence at the lower boundary, a different quantity, which they note “has little impact on the overall atmospheric structure”.) Lampón et al. (2020, 2021) is the reference implementation, with turbulent broadening as a free or assumed parameter of 1D isothermal Parker fits; Czesla et al. (2022) follows it. Cherubim et al. (2026) inherits the choice through p-winds’ `turbulence_broadening=True`, so the LHS 1140 b retrieval carries it — a fact stated nowhere in the paper text.

Test it, then leave it out. Yan et al. (2024) give the cleanest statement of the convention and of the doubt. They record that “these authors assumed that the turbulence velocity $v_{\text{turb}} = \sqrt{5kT/3m}$, which corresponds to the speed of sound of a monatomic gas and, thus, to the transonic case with a Mach number of 1”, then keep it out of their nominal models and test $v_{\text{turb}} = 0.5, 1.0$ and $1.5 c_s$ as a sensitivity. Their finding for He 10830 is that turbulence “decreases the absorption depth slightly at the line center, but doesn’t significantly alter the absorption at the line wings”. They argue Mach 1 is likely an overestimate, since in the diffuse ISM turbulent velocity anti-correlates with temperature and the Mach number is below unity at a few thousand K, and conclude “the turbulence effect is likely less significant than the one explored in this paper”. Yan et al. (2019) computes the Doppler width explicitly “assuming no turbulence”.

Argue against it. Murray-Clay et al. (2009), on García Muñoz’s (2007) proposal to broaden $\text{Ly}\alpha$ by wind turbulence: “to generate the large velocities observed, the energy in such turbulence would need to exceed the thermal and bulk kinetic energies in the mean flow by a factor of ~ 100 . Such energy requirements seem insurmountable.” Koskinen et al. (2013b) find that “non-thermal broadening such as that proposed by Ben-Jaffel and Hosseini (2010) does not appear to be necessary to explain the current observations”, while allowing that stellar-wind interaction may produce broadening turbulence. Schulik & Owen (2025) offer a physical alternative — adiabatic cooling produces a metastable-helium population bump at high velocity, widening the line with no free parameter — and are pointed about the alternative: the isothermal model “can only explain the broadness of the data under physically inconsistent

assumptions, i.e. those of even larger mass-loss rates, not supported by the energy input into this planet, and an empirical free parameter, turbulent broadening (Lampón et al. 2021)”. They add that $T = 10^4$ K already maximizes what thermal broadening alone can give.

Need it even so. Taylor et al. (2026) is the closest analogue to the present work: a self-consistent model (energy balance, multispecies diffusion, non-LTE $H(n=2)$) that reproduces the HD 189733 b line core but not its width. They write that “several prior analyses have matched the HD 189733 b width by introducing ad hoc (micro)turbulent or Gaussian broadening terms; here, we show that this step remains necessary even in our self-consistent framework”, and quantify it as $\sim 12 \text{ km s}^{-1}$ of additional non-thermal broadening. They speculate — explicitly calling it speculative — that it is plasma turbulence from ions trapped on low-latitude planetary field lines, strengthening with stellar activity. Rumenskikh et al. (2022), as reported there, instead imposed reduced hydrogen line cooling and prescribed zonal jets of $10\text{--}20 \text{ km s}^{-1}$ in a 3D multifluid model.

Matches that are not about broadening. Recorded so the sweep need not be repeated: Wogan (2025), Lavvas (2014, 2019), Robeling (2026) and the second half of Koskinen et al. (2013b) use “turbulent” for eddy diffusion K_{zz} ; 2607.18193v1 for flow–flow interaction in a 3D stellar-wind simulation; Jensen (2012) for photospheric turbulence driving stellar activity; Taylor et al. (2025) for gravity-wave breaking. Vidal_2003_Nature.pdf matches only on its first page, which carries the tail of the preceding *Nature* letter (on Crab pulsar radio emission); Vidal-Madjar et al. themselves do not discuss turbulence.

Where that leaves this comparison. There is no settled treatment. Turbulence at $v_{\text{turb}} = c_s$ is a common convention in 1D isothermal Parker-wind fits — and hence in the paper being reproduced — while self-consistent and 3D models either need a larger ad hoc term, reach the width by other physics, or argue the term is unjustified. Enabling it here is therefore right for a like-for-like comparison and wrong as a default, which is how it is implemented.

5. Searching for a match: the composition scan

Once the star, the planet and the spectrum are fixed, composition is the free parameter a match has to be sought in first. This section takes it in the well-mixed limit — element diffusion off, so the reservoir ratio is also the ratio everywhere in the wind. That is the simpler of the two mixing assumptions available, and Sect. 8.3 repeats the whole exercise with binary H/He diffusion active, where the answer depends on the eddy coefficient as well.

Eleven compositions were solved, from solar ($\text{He}/\text{H} = 0.083$) to $\text{He}/\text{H} = 10^4$, each a JFNK steady state ($\|R\| < 10^{-3}$, info = 0; one case, $\text{He}/\text{H} = 100$, stagnates at 1.18×10^{-3} and is recorded as such). Nothing is imposed: mass, momentum and energy are solved with photoionization heating and radiative cooling, so $\dot{M}$ and $T(r)$ are outputs.

He/H	$\log_{10} \dot{M}$ [g s ⁻¹]	red [%]	red (turb) [%]	FWHM [Å]	EW [% Å]
0.083	7.77	1.21	1.08	0.218	0.279
0.25	7.75	2.65	2.32	0.230	0.645
0.50	7.76	4.05	3.50	0.242	1.040
0.55	7.76	4.29	3.69	0.244	1.109
0.70	7.78	4.97	4.26	0.248	1.311
1	7.80	5.88	5.01	0.255	1.598
10	7.76	7.42	6.14	0.287	2.270
10 ³	7.74	5.67	4.66	0.304	1.844
10 ⁴	7.73	5.60	4.60	0.304	1.825
observed	—	1.254	—	0.841	1.108 ± 0.030

Nine of the eleven runs are listed; He/H = 0.6 falls between two tabulated entries and He/H = 100 is the case whose residual stagnates. The three helium-dominated rows are corrected values: their depths and equivalent widths were an artifact of the old advection gate, which pinned the metastable population to its equilibrium value across the hydrogen ionization front, and they fell by 89–93% when the gate was rebuilt on the slowest solved species (Update_EXHALE.md Sect. 72; at He/H = 10, red 75.97% → 7.42% and EW 21.53 → 2.270 % Å). The crossing of Sect. 7. is interpolated inside He/H ≤ 1 and is untouched by that correction.

The escape rate is nearly composition-independent, $\dot{M} \simeq 6 \times 10^7 \text{ g s}^{-1}$, about a third of the 2.03×10^8 the retrieval imposes.

6. Why the two models disagree

At the same composition the two models are not small perturbations of each other. The *total* metastable column is comparable — at H:He = 10⁻³ EXHALE’s is about half of p-winds’ — but its distribution is not, and transit depth is absorbing *area*:

	p-winds	EXHALE
50% of the 2 ³ S column inside	1.33 R_p	5.7 R_p
90% inside	1.59 R_p	10.2 R_p

Both radii are read from the local-equilibrium metastable profile, which is what Fig. 2 draws. The advection-corrected profile that the transit integral consumes is more concentrated, and for this helium-dominated composition it collapses onto the base once the advection gate is formed on the metastable’s own clock rather than on hydrogen’s (Update_EXHALE.md Sect. 72). This section is therefore a statement about the two wind solutions; every transit number quoted elsewhere in the memo is measured on the advection-corrected profiles.

The star is 13.6 R_p across, so metastable helium spread to 10 R_p covers most of the disk while a column confined to 1.6 R_p can darken only a narrow ring. The distribution difference is the wind structure — the one thing p-winds imposes

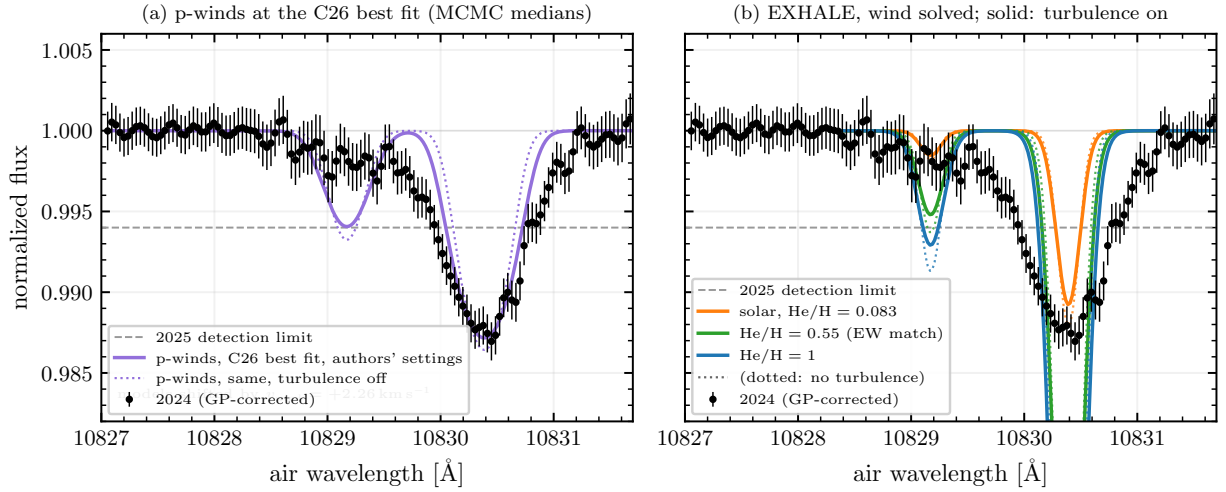


Figure 1: The measurement (black) with, left, the paper’s own p-winds model at its best fit — the MCMC medians its Fig. 4 legend states — computed with the authors’ released settings (solid; dotted is the same run with turbulence off), and right, EXHALE at three compositions. Solid EXHALE curves carry the same turbulence term p-winds uses; dotted are without it. Neither code applies a line-of-sight bulk velocity, so both sets of model curves are shifted by the $v_{\text{wind}} = +2.26 \text{ km s}^{-1}$ that C26 fit; the shift is rigid and leaves equivalent widths unchanged. Drawn in the presentation of the paper’s Fig. 4.

and EXHALE solves. At $r = 10R_p$, same composition and spectrum: p-winds is isothermal at 6100 K by construction, EXHALE’s energy balance gives 1392 K; p-winds flows at 4.3 km s^{-1} , EXHALE at 0.53; and the cold, slow, denser gas keeps helium neutral and the metastable level populated far out, $n(2^3S) = 40 \text{ cm}^{-3}$ against 2×10^{-4} (Fig. 2).

7. What can be matched: the equivalent width

Depth and width trade against each other under any broadening, but the equivalent width does not: it is invariant under the instrument profile, the turbulence term and the phase averaging. It is therefore the quantity on which a self-consistent model can be matched without first settling the broadening question.

Over the blended red pair (vacuum 10832.6–10834.2 Å), the released spectrum gives $EW = 1.108 \pm 0.030 \text{ \% Å}$. EXHALE crosses it at

$$\text{He/H} \simeq 0.55 \quad (0.53\text{--}0.57 \text{ on the } EW \text{ } 1\sigma), \quad \text{H:He} \simeq 1.8, \quad (3)$$

that is $6.6\times$ solar, a helium mass fraction of 0.69. Enabling turbulence moves this to 0.541, a 1.6% shift — a direct demonstration that the estimate does not depend on the broadening treatment. The value sits three orders of magnitude away from the paper’s $\text{H:He} \lesssim 10^{-3}$, and the reason is the structural difference of the previous section: a wind that spreads metastable helium over most of the stellar disk needs far less helium to absorb as much.

Two assumptions are carried inside that number and are taken up later. It is the well-mixed crossing, and it is the crossing on the GJ 1132 proxy spectrum. Sect. 8.3 rescans it with element diffusion active at every decade of the eddy

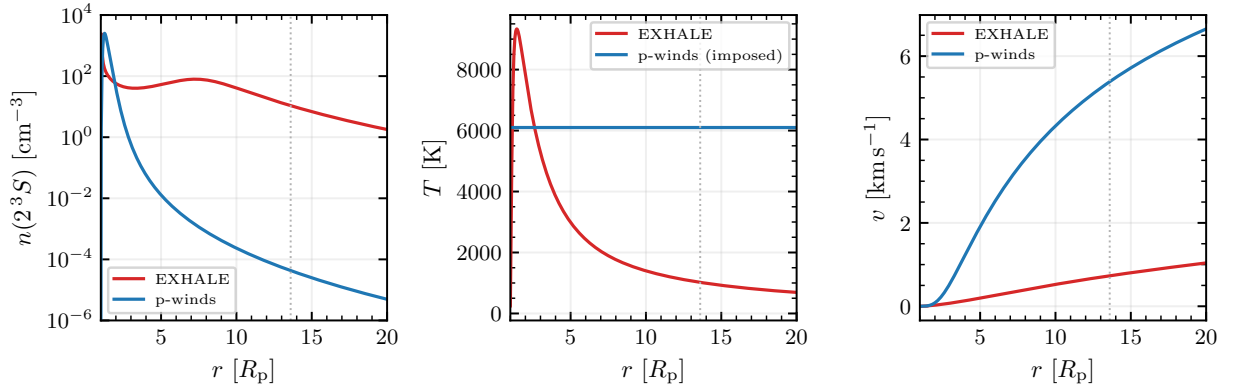


Figure 2: Same composition ($\text{H}:\text{He} = 10^{-3}$), same spectrum, two wind structures. Dotted vertical line: the stellar radius, $13.6R_p$.

coefficient, and Sect. 10. rescans it on the brighter proxy; both move it, the spectrum by much the larger factor.

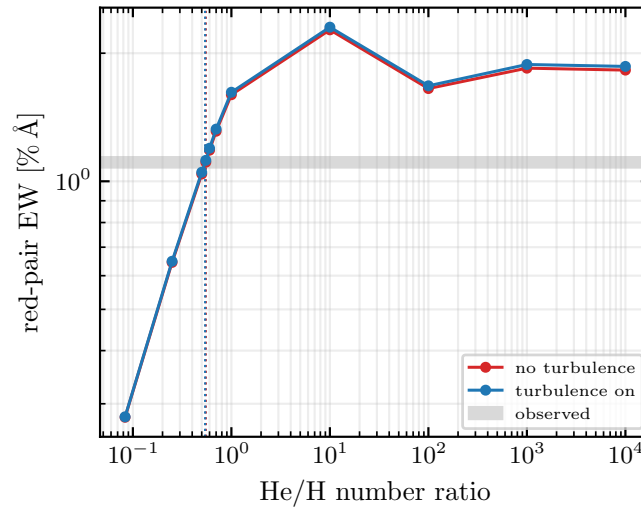


Figure 3: Red-pair equivalent width against composition, with and without the turbulence term, against the measured band. The crossing is insensitive to the broadening treatment.

8. What cannot be matched: the profile

Matching the equivalent width is not matching the line. At $\text{He}/\text{H} = 0.55$ the profile is 3.4 times too deep (4.29% against 1.254%) and 3.4 times too narrow (0.244 against 0.841 Å). The equivalent width states how much absorbing power the model has; the profile states how that power is spread in velocity, and composition does not act on the second. Every solved composition gives 0.22–0.30 Å, rising only to 0.24–0.38 Å with the turbulence term.

8.1 How much is missing, in velocity

The measured FWHM is $0.841 \text{ \AA}$, that is 23.3 km s^{-1} at $10833 \text{ \AA}$. Three ways to supply it, and what each would require:

- *Thermal alone.* A Doppler width of 23.3 km s^{-1} FWHM in helium needs $T = 4.7 \times 10^4 \text{ K}$. Helium is fully ionized far below that, so there would be no metastable population left to absorb. This is excluded.
- *Bulk outflow.* A line-of-sight velocity spread of order $\pm 12 \text{ km s}^{-1}$ across the line-forming region would do it. The EW-matched EXHALE solution reaches 1.2 km s^{-1} at $20 R_p$; the paper's p-winds model, with its imposed temperature, reaches 4–7. Both are an order of magnitude short.
- *Non-thermal (turbulent) broadening.* Added in quadrature to what EXHALE already has, closing the gap takes 22.3 km s^{-1} FWHM — 9.5 km s^{-1} of turbulent velocity (σ), against the 2.9 km s^{-1} that the sound-speed convention of Sect. 4.1 supplies for helium at 5000 K. This is the one of the three that can be measured directly, and Sect. 8.2 measures it.

So the model is not short by a detail. It is short by an amount of velocity comparable to the escape speed, and no choice of composition supplies it.

8.2 The broadening the data demand, measured

The three routes above can be replaced by one measurement. Take the EW-matched EXHALE profile as solved, convolve it with a Gaussian of FWHM f in velocity, and ask which f reproduces the measured line. The convolution is applied to the excess absorption $(\max T - T) / \max T$ built from the instrument-convolved column of `tpm_He10830.txt`; depth, width and the red/blue ratio are then read off with the same three-Gaussian extractor applied to the data, and the equivalent width by the same integral over the vacuum window $10832.60\text{--}10834.20 \text{ \AA}$. So f is the only quantity varied, and every number in Table 1 is measured the way the corresponding observed number was.

Table 1: The EW-matched EXHALE profile ($\text{He}/\text{H} = 0.55$, no turbulence term) convolved with an added Gaussian of FWHM f . Observed values on the last row: red and blue depths from the C26 MCMC posterior, width and equivalent width measured here on their spectrum.

$f \text{ [km s}^{-1}\text{]}$	red [%]	blue [%]	red/blue	FWHM [$\text{\AA}$]	EW [% $\text{\AA}$]
0	4.285	0.624	6.86	0.244	1.109
10	2.393	0.321	7.45	0.435	1.109
15	1.754	0.232	7.56	0.593	1.106
20	1.366	0.180	7.61	0.762	1.090
22.3	1.237	0.162	7.62	0.841	1.076
25	1.114	0.146	7.63	0.935	1.056
observed	$1.24^{+0.22}_{-0.23}$	$0.25^{+0.14}_{-0.12}$	$6.7^{+12.7}_{-3.1}$	0.841	1.108 ± 0.030

The width is reached at $f = 22.3 \text{ km s}^{-1}$ FWHM, that is $\sigma = 9.5 \text{ km s}^{-1}$. At that one value the depth is 1.237% against the measured $1.24^{+0.22}_{-0.23}$, the blue component 0.162% against $0.25^{+0.14}_{-0.12}$, and the ratio 7.62 against $6.7^{+12.7}_{-3.1}$: depth, blue

depth and ratio all land inside the observed 1σ interval at the value fixed by the width alone. Nothing was tuned to them. Figure 4 shows the profile and the two crossings.

The number does not depend on which solution is broadened. The same crossing is at 21.9 km s^{-1} for the $\text{He}/\text{H} = 0.55$ run with the turbulence term already on, and at 22.5 km s^{-1} for solar composition — although the solar run is a quarter of the observed absorption (0.311% red, EW 0.270) and so matches the shape while missing the amount. The required broadening is therefore set by the velocity field, not by composition or by the broadening treatment already in place: the direct measurement of what Sect. 8. inferred. The equivalent width falls by 3% between $f = 0$ and $f = 22.3$ because the Gaussian wings carry absorption out of the red-pair integration window, not because absorbing power is lost.

This is descriptive, not explanatory. It states that one free parameter, applied to a solved profile, reproduces the measured line; it does not say what produces 22.3 km s^{-1} . The candidates are the ones listed below, and the measurement sharpens them by fixing the number each has to deliver.

One conversion narrows the list before it is read. If the missing 22.3 km s^{-1} is called turbulence, it has to be turbulence in gas whose sound speed the solution itself fixes. Evaluating $c_s = (\gamma kT / \mu m_H)^{1/2}$ with $\gamma = 5/3$ on the $\text{He}/\text{H} = 0.55$ output, taking μ from the solved H I/H II/He I/He II/He III and electron densities, gives $c_s = 5.5 \text{ km s}^{-1}$ at $2R_p$ ($T = 4311 \text{ K}$, $\mu = 1.94$), 3.8 at the metastable peak $5.4R_p$ (1609 K, 1.57) and 3.0 at $10R_p$ (845 K, 1.32); weighted over the line-forming region by $n(2^3S)r$, $T = 2022 \text{ K}$, $\mu = 1.59$ and $c_s = 3.9 \text{ km s}^{-1}$. Note this is the bulk-gas sound speed, the quantity a Mach number is defined against, and not the helium thermal speed that enters the line profile.

Against that, the kernel is supersonic on any convention. Read as a line-of-sight velocity dispersion, $\sigma = 9.5 \text{ km s}^{-1}$ is $\mathcal{M} = 2.4$. Read in the b -parameter convention of Lampón et al. and p-winds, where v_{turb} is the quantity added to $b^2 = 2kT/m + v_{\text{turb}}^2$, it is $v_{\text{turb}} = \sqrt{2}\sigma = 13.4 \text{ km s}^{-1}$, $\mathcal{M} = 3.4$; as an isotropic three-dimensional turbulent speed, $\sqrt{3}\sigma = 16.4 \text{ km s}^{-1}$, $\mathcal{M} = 4.2$. The $v_{\text{turb}} = c_s$ convention of Sect. 4.1 supplies only 3–4 km s^{-1} in this gas, about a third of what the data demand. That is the opposite corner of the literature reviewed there: Yan et al. (2024) argue $\mathcal{M} = 1$ is already an overestimate at a few thousand K, and Murray-Clay et al. (2009) call the energy requirement of supersonic turbulence “insurmountable”. Supplying the shortfall under the name of turbulence therefore looks physically hard to sustain, and the measurement points to the candidates of the next section — an ordered velocity field, three-dimensional flow, or heating the 1D energy equation does not carry — rather than to a missing thermal width.

8.3 The same demand, under diffusion and the adopted K_{zz}

Everything above was solved in the well-mixed limit. With binary H/He element diffusion active, helium settles out of the base unless an eddy term lifts it, and how much of it reaches the line-forming region is set by the eddy coefficient K_{zz} , which is not measured for this system. The diffusion operator is described in `binary_diffusion_design.md`; the literature range at the $1 \mu\text{bar}$ base of these runs, the ladder of solved cases, and the adoption of $K_{zz} = 1.0 \times 10^9 \text{ cm}^2 \text{ s}^{-1}$ are in `LHS1140b/kzz_decision.md`.

The composition, and how much of it K_{zz} carries

Turning the operator on moves the equivalent-width crossing of Fig. 3 away from $\text{He}/\text{H} = 0.55$, and by how much depends on K_{zz} . The composition was therefore rescanned at every decade of the eddy coefficient from $K_{zz} = 0$ to

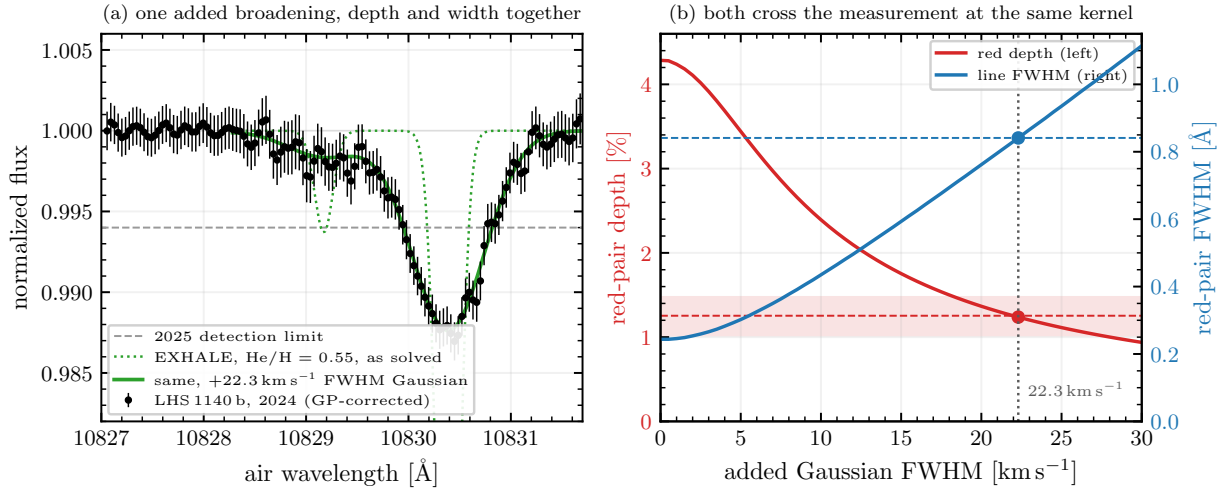


Figure 4: Left: the EW-matched EXHALE profile as solved (dotted, its core running below the plotted range) and convolved with a 22.3 km s^{-1} FWHM Gaussian (solid), against the measurement. Both are shifted by the same $v_{\text{wind}} = +2.26 \text{ km s}^{-1}$ used in the earlier figures. Right: red-pair depth (red, left axis) and red-pair FWHM (blue, right axis) as the added kernel is scanned. Dashed lines are the observed values, the shaded band the C26 1σ interval on the depth; the dotted vertical marks where the width is matched. Both curves meet the measurement at the same kernel.

$10^{11} \text{ cm}^2 \text{ s}^{-1}$, by the same route in every case: the same control input, the same direct-steady JFNK solve at $\|R\|_{\text{tol}} = 5.0 \times 10^{-3}$, the same single post-processing pass and transit synthesis, and the same red-pair equivalent width on the vacuum window $10832.60\text{--}10834.20 \text{ Å}$. Table 2 is the result; every crossing is interpolated inside a bracket of two solved runs, none extrapolated and none against a saturated line — at $K_{zz} = 10^6$, where the crossing sits highest, the equivalent width is still climbing at $\text{He}/\text{H} = 10$.

The full ladder is three regimes, not one power law (Fig. 5). Above 10^7 the degeneracy is shallow and very nearly a power law, $d \log(\text{He}/\text{H})/d \log K_{zz} = -0.136$ ($-0.139, -0.133$ and -0.141 across the three decades $10^8\text{--}10^{11}$), so $\text{He}/\text{H} \propto K_{zz}^{-0.14}$. Below 10^7 the curve rolls over (-0.071 across $10^6\text{--}10^7$, -0.022 across $10^5\text{--}10^6$) onto a plateau at $\text{He}/\text{H} = 4.68$, the value the line implies when the eddy term does nothing. That plateau is where it is for a reason that is measured rather than assumed: the operator adds K_{zz} to the molecular coefficient, and these helium-rich solutions carry $D_{\text{eff}} = 1.24\text{--}1.32 \times 10^6 \text{ cm}^2 \text{ s}^{-1}$ at the base, so $K_{zz} = 10^6$ raises the base coefficient by 77%, 10^5 by 8% and 10^4 by 0.8%. At a fixed $\text{He}/\text{H} = 5$ the resulting equivalent width departs from the $K_{zz} = 0$ one by 0.005 of the measurement error at 10^4 , 0.18 at 10^5 and 1.76 at 10^6 : *no eddy coefficient at or below $10^4 \text{ cm}^2 \text{ s}^{-1}$ is distinguishable from none at all by this line*. At the other end the operator never reaches the well-mixed limit either — even at 10^{11} the inferred reservoir, 1.11, is twice the well-mixed 0.55.

Over the span the literature leaves open at the base pressure of these runs the inferred reservoir ratio moves only from 1.5 to 2.9, a factor 1.9, against a factor 9 in equivalent width across the same two decades at fixed composition: the line responds to composition much faster than it responds to K_{zz} , so a large change in the eddy coefficient is bought back by a small change in the reservoir. *The composition inferred from this line under diffusion is therefore only weakly conditional on K_{zz} , as $K_{zz}^{-0.14}$, and strongly conditional on the assumed spectrum* — the SED moves the well-mixed

Table 2: Where the red-pair equivalent width crosses the measured $1.108 \pm 0.030 \text{ \% } \text{\AA}$, at every decade of the eddy coefficient, with binary H/He diffusion active throughout. The 1σ range propagates the measurement error alone, within one K_{zz} . The decades 10^1 – 10^4 were not scanned in composition: at a fixed He/H = 5 their equivalent width differs from the $K_{zz} = 0$ one by 0.00002 to 0.005 of the measurement error, which is the entry they carry. Runs, seeds and residuals: LHS1140b/kzz_decision.md, Sect. 6.1.

$K_{zz} [\text{cm}^2 \text{s}^{-1}]$	He/H at the crossing	1σ range	bracketing runs [EW in $\text{\AA}$]
0 (molecular only)	4.68	4.54–4.83	4.6 [1.092], 4.8 [1.134]
10^1 – 10^4	not distinguishable from $K_{zz} = 0$ by this line		
10^5	4.68	4.55–4.83	4.6 [1.092], 4.8 [1.133]
10^6	4.45	4.33–4.59	4.4 [1.096], 4.7 [1.161]
10^7	3.78	3.66–3.96	3.6 [1.063], 3.8 [1.113]
10^8	2.87	2.78–2.97	2.70 [1.050], 3.00 [1.149]
10^9 (adopted)	2.09	2.03–2.16	2.00 [1.065], 2.13 [1.129]
10^{10}	1.54	1.47–1.59	1.50 [1.087], 1.60 [1.146]
10^{11}	1.11	1.07–1.15	1.00 [1.019], 1.20 [1.180]
none (well mixed)	0.550	—	Fig. 3

crossing by a factor 9, from 0.550 to 0.060 (Sect. 10.), which is five times the entire K_{zz} lever.

What K_{zz} does act on strongly is the helium profile itself, and Fig. 6 separates that from the wind. At a fixed reservoir He/H = 0.55, raising K_{zz} from 0 to 10^{10} lifts the homopause from below the base to $1.13 R_p$ and the metastable population by a factor 400, while the velocity and density profiles move by less than 30% and $\dot{M}$ by 0.23 dex. Temperature is the one bulk variable that follows, because the helium the eddy term lifts is itself a heating agent. Along the equivalent-width-matched sequence of Table 2, where each K_{zz} carries its own crossing composition, $\log_{10} \dot{M} = 7.80 \pm 0.01$ in every run: the escape rate does not carry the K_{zz} dependence at all.

The width the data demand, again

The question that remains is narrower: whether the width the data demand moves with the composition.

It does not. Repeating the measurement of Sect. 8.2 on heh2p13_diff_kzz1e9, the EW-matched diffusive solution (He/H = 2.13, red-pair EW $1.129 \text{ \% } \text{\AA}$, red depth 4.396%, FWHM $0.242 \text{ \AA}$), with the same convolution, the same three-Gaussian extractor and the same integration window, gives Table 3.

The width is reached at $f = 22.31 \text{ km s}^{-1}$ FWHM, against 22.30 km s^{-1} for the diffusion-off He/H = 0.55 solution of Sect. 8.2: the two differ by 0.01 km s^{-1} , more than three orders of magnitude below the number itself. At that kernel the depth is 1.260% against the measured $1.24^{+0.22}_{-0.23}$, the blue component 0.166% against $0.25^{+0.14}_{-0.12}$, the ratio 7.60 against $6.7^{+12.7}_{-3.1}$ and the equivalent width 1.094 against 1.108 ± 0.030 — all four inside the observed 1σ interval at the value fixed by the width alone, as in the well-mixed case. So the velocity the profile demands is unchanged by turning diffusion on and by the adopted eddy coefficient: those change how much helium is there, not how fast it moves.

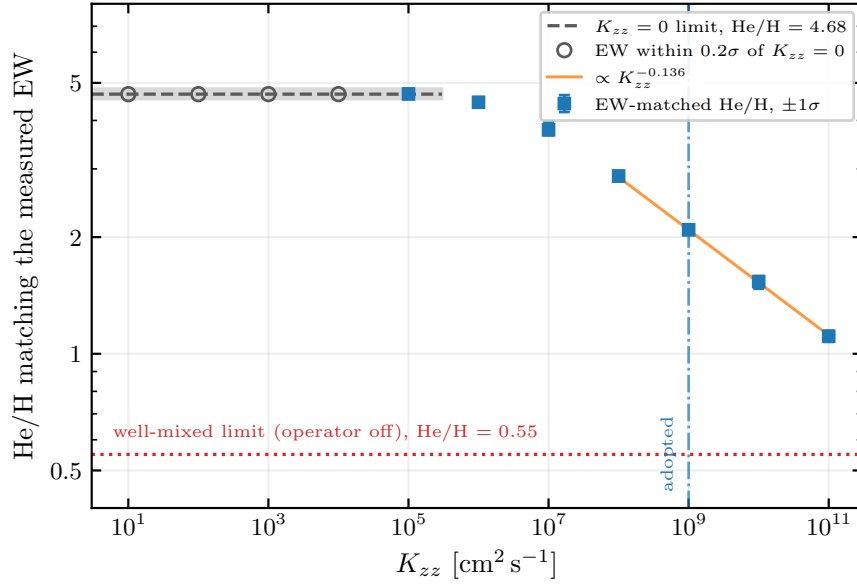


Figure 5: **The composition–mixing degeneracy in full.** Each square is the He/H reservoir ratio whose red-pair equivalent width equals the measured one, solved at one decade of K_{zz} with binary H/He diffusion active (Table 2); the error bars propagate the measurement error alone and are smaller than the symbols over most of the range. The landscape has three parts. Below $K_{zz} \simeq 10^5$ the eddy term is a small correction to the molecular coefficient the base already carries, the inference sits on a plateau at He/H = 4.68 (dashed, with its 1σ band), and the open circles mark the decades whose equivalent width differs from the $K_{zz} = 0$ one by less than a fifth of the measurement error. Between 10^6 and 10^7 the curve rolls off. From 10^8 up it is the shallow power law $K_{zz}^{-0.136}$ (orange). Neither end of the axis reaches the well-mixed crossing of 0.55 (dotted): with the operator active, helium settles at every eddy coefficient tried, and He/H inferred from this line is a statement about the assumed mixing as much as about the planet. The adopted $K_{zz} = 10^9 \text{ cm}^2 \text{ s}^{-1}$ is marked.

Table 3: The EW-matched diffusive solution (heh2p13_diff_kzz1e9: He/H = 2.13, $K_{zz} = 10^9 \text{ cm}^2 \text{ s}^{-1}$) convolved with an added Gaussian of FWHM f , measured exactly as Table 1.

f [km s ⁻¹]	red [%]	blue [%]	red/blue	FWHM [Å]	EW [% Å]
0	4.396	0.643	6.84	0.242	1.129
10	2.441	0.328	7.44	0.434	1.129
15	1.787	0.237	7.55	0.593	1.125
20	1.392	0.183	7.59	0.761	1.108
22.3	1.260	0.166	7.60	0.841	1.094
25	1.134	0.149	7.62	0.935	1.073
observed	$1.24^{+0.22}_{-0.23}$	$0.25^{+0.14}_{-0.12}$	$6.7^{+12.7}_{-3.1}$	0.841	1.108 ± 0.030

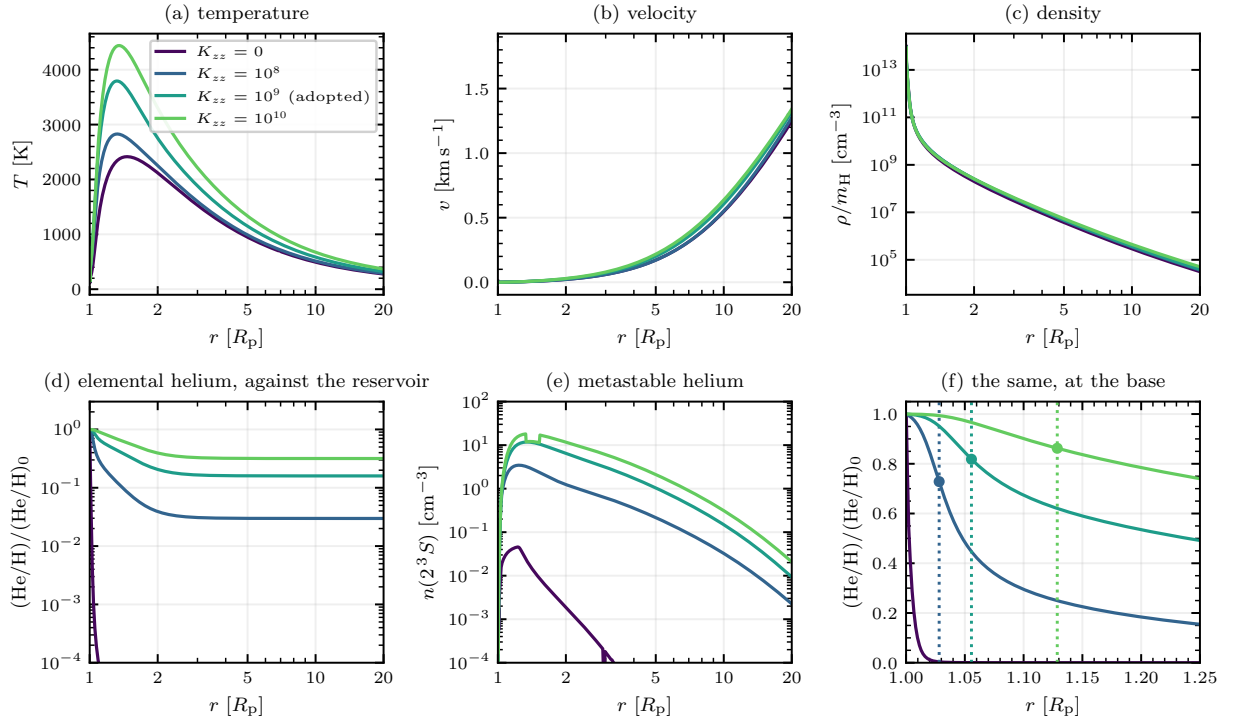


Figure 6: Four converged solutions differing only in the eddy coefficient, at a fixed reservoir $\text{He}/\text{H} = 0.55$ with binary H/He diffusion active: $K_{zz} = 0, 10^8, 10^9$ (adopted) and $10^{10} \text{ cm}^2 \text{ s}^{-1}$, dark to light. **The eddy coefficient is a composition knob, not a wind knob.** Panels (d)–(f), the elemental helium against its base reservoir value and the metastable population, span two to four orders of magnitude across the four runs — at $1.05 R_p$ the elemental ratio is 0.000, 0.485, 0.845, 0.974, and the peak $n(2^3S)$ is 0.046, 3.5, 11.8, 18.1 cm^{-3} — while in panels (b) and (c) the velocity at $10 R_p$ moves by 16% and the density at $2 R_p$ by 28%, and $\log_{10} \dot{M}$ by 0.23 dex. Panel (a) is the exception: T at $1.5 R_p$ rises from 2412 to 4276 K, because the helium the eddy term lifts is itself absorbing. Panel (f) repeats (d) on a linear axis over the base, where the dots and dotted lines mark each run’s homopause — the radius at which its own molecular coefficient equals its K_{zz} — at 1.028, 1.056 and 1.129 R_p , with the $K_{zz} = 0$ homopause below the base.

Figure 7 shows the two broadened curves lying on each other.

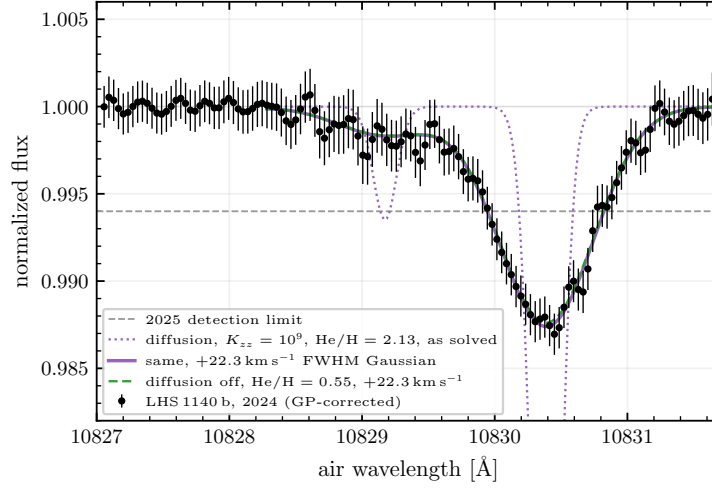


Figure 7: The EW-matched diffusive solution as solved (dotted purple, its core running below the plotted range) and convolved with the 22.3 km s^{-1} FWHM Gaussian that reproduces the measured width (solid purple), against the measurement; the diffusion-off $\text{He}/\text{H} = 0.55$ curve at its own matched kernel (dashed green) is repeated from Fig. 4. The two broadened curves are indistinguishable — their kernels differ by 0.01 km s^{-1} . All curves are shifted by the same $v_{\text{wind}} = +2.26 \text{ km s}^{-1}$.

8.4 Broadening cannot repair a deficit in equivalent width

The same scan was applied to the p-winds curves, in their own vacuum frame and with the same three-Gaussian extractor and the same integration window (Table 4). The convolution conserves absorbing power: it redistributes the excess in velocity and leaves the equivalent width unchanged, apart from the few percent that the wings carry out of the integration window. Whether one kernel can reproduce depth and width together is therefore decided before the scan starts, by the equivalent width the curve already has at $f = 0$.

Table 4: The same added-Gaussian scan applied to the p-winds curves, with the EXHALE row of Table 1 repeated for comparison. The kernel is the one that reproduces the measured width, 0.841 Å ; the depths and the ratio are read at that kernel. Equivalent widths are for the unbroadened curve, over the vacuum window $10832.60\text{--}10834.20 \text{ Å}$.

curve	EW($f=0$) [% Å]	f [km s^{-1}]	red [%]	blue [%]	red/blue
p-winds best fit, turbulence on	0.837	17.8	0.926	0.339	2.73
p-winds best fit, turbulence off	0.701	20.1	0.777	0.301	2.58
p-winds depth-matched, 6100 K	0.480	20.2	0.535	0.140	3.82
EXHALE, $\text{He}/\text{H} = 0.55$	1.109	22.3	1.237	0.162	7.62
observed	1.108 ± 0.030	—	$1.24^{+0.22}_{-0.23}$	$0.25^{+0.14}_{-0.12}$	$6.7^{+12.7}_{-3.1}$

The p-winds best fit carries $0.837 \% \text{ Å}$ against the measured 1.108 ± 0.030 , 24% short. At the kernel that reproduces the

width, $f = 17.8 \text{ km s}^{-1}$, the red depth is 0.926% against $1.24^{+0.22}_{-0.23}$, about 1.4σ low. What the broadening does is move the disagreement rather than remove it: unbroadened, the curve is too deep in the blue component (0.534% against $0.25^{+0.14}_{-0.12}$, some 2σ high) and too narrow; broadened to the measured width, the blue component falls to 0.339% , inside 1σ , and the red component drops below the measurement. The blue side is genuinely improved; the red side is not, and the total is fixed.

The intensity ratio barely moves, 2.57 at $f = 0$ against 2.73 at the matched kernel. It is the amplitude ratio of a saturated pair, set by how the two components share the optical depth, and a common convolution acts on both alike. It stays well below the measured $6.7^{+12.7}_{-3.1}$, though the observed interval is wide enough that this alone is not a discriminator.

Matching the depth as well would require 24% more absorbing power, and that is not something a kernel supplies: it is a change in $\dot{M}$, temperature or composition. Fitting the broadening as a free parameter alongside those — which is what the measurement asks for — would presumably shift the C26 parameter estimates, since the composition is constrained through the same absorbing power that would have to rise.

The asymmetry with EXHALE should not be read as one code outperforming the other. The EXHALE curve entered the scan with its equivalent width already set to the observed value by the choice of $\text{He}/\text{H} = 0.55$ (Sect. 8.2), so only the shape remained free, and one kernel settled it. The p-winds curve entered at the composition C26 report, whose equivalent width falls short, so no kernel could settle both. The two scans test the same physical statement — broadening redistributes, it does not create — from two different starting points.

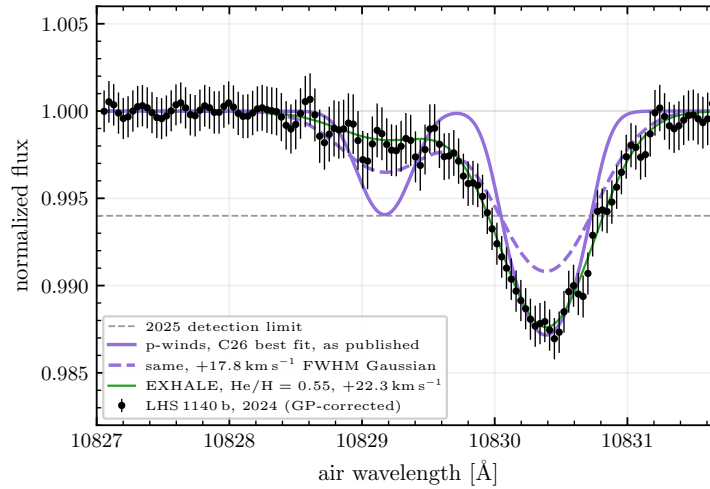


Figure 8: The C26 p-winds best fit as published (solid purple) and convolved with the 17.8 km s^{-1} FWHM Gaussian that reproduces the measured width (dashed purple), against the measurement; the EXHALE $\text{He}/\text{H} = 0.55$ curve at its own matched kernel (22.3 km s^{-1} , thin green) is repeated from Fig. 4. Broadened to the observed width, the p-winds curve passes above the red core: the width is matched and the depth is not. All curves are shifted by the same $v_{\text{wind}} = +2.26 \text{ km s}^{-1}$.

8.5 The adiabatic-cooling population bump, looked for

Schulik & Owen (2025) propose that the missing width is already contained in the metastable population itself. Writing the steady state of their Eq. (1) as

$$n_{2^3S}(T_e) = \frac{\alpha(T_e)}{q(T_e) + t_{\text{ion}}/n_e} n_{\text{He}^+}, \quad (4)$$

they note that as adiabatic expansion drops the temperature “from $\simeq 10^4$ K to $\simeq 10^3$ K”, “a change in slope in the population occurs as the dominant triplet loss rate $q(T_e)$ drops significantly at lower temperatures”—the metastable fraction climbs above the curve computed with the rates frozen at the peak temperature. The rise is “quenched” where the advection timescale overtakes the excitation to 2^1S , after which $\dot{m}_{2^3S} = 4\pi r^2 v n_{2^3S} m_{\text{He}}$ falls through ionization and the $2^3S \rightarrow 1^1S$ decay. They credit Yan et al. (2022) and Biassoni et al. (2024) with first identifying the effect, and their conclusion states the condition it must satisfy to be observable: “The helium triplet population bump must be situated at high velocities to help explain anomalous line widths.”

EXHALE carries the temperature dependence the mechanism needs, and carries it in exactly the balance above. The metastable row (`tr_triplet_row`, `ion_residual_core.f90:72`) is

$$n_e (n_{\text{He}^+} \alpha_{2^3S} + n_{\text{HeI}(1^1S)} q_{13}) = n_{2^3S} [\Gamma_{2^3S} + A_{31} + n_{\text{H1}} Q_{31} + n_e (q_{31a} + q_{31b} + b_{2^3S})], \quad (5)$$

with every coefficient a function of T : $\alpha_{2^3S} = 2.10 \times 10^{-13} (T/10^4)^{-0.778}$ (`Cool_coeff.f90:970`), the collisional losses q_{31a} ($2^3S \rightarrow 2^1S$, 0.80 eV) and q_{31b} ($2^3S \rightarrow 2^1P$, 1.40 eV) as Bray (2000) collision strengths folded with $\exp(-\Delta E/kT)$ (`Cool_coeff.f90:905` and `928`), the Penning term Q_{31} as the Taylor et al. (2025) two-branch fit (`Cool_coeff.f90:988`), collisional ionization of the metastable (`Cool_coeff.f90:1030`), and the radiative decay $A_{31} = 1.272 \times 10^{-4} \text{ s}^{-1}$ (`util_ion_eq.f90:1192`). Between 5000 and 1000 K the loss $q_{31a} + q_{31b}$ falls by a factor 1.3×10^3 while α_{2^3S} rises by a factor 3.5, so the ingredients of the bump are present and are not an imposed parameterization.

LHS1140b/bump_analysis.py measures the result. It reconstructs the balance above in Python from the solved profiles—reproducing the solved n_{2^3S} to a relative 3×10^{-16} , which confirms the rate transcription and lets the one unresolved term, the 2^3S photoionization rate Γ_{2^3S} , be recovered from the solution itself—and then recomputes the reference curve with α , q_{13} , q_{31a} , q_{31b} and Q_{31} frozen at the peak temperature.

The bump is there. In every case the metastable fraction $f_3 = n_{2^3S}/n_{\text{He}}$ rises monotonically outward across the whole cooling leg, from the temperature maximum to a maximum of f_3 at $1.1\text{--}16.2 R_p$ —a rise of $1.4\text{--}4.2 \times 10^3$ in the equilibrium profiles and $22\text{--}236$ in the advection-corrected ones—and the rise is attributable to the temperature dependence in the way Schulik & Owen describe: against the same balance with the rates frozen at the peak temperature, the solved population stands a factor 1.6–31 high at the density maximum and up to 8.8–156 high further out. On the GJ 1132 proxy at the equivalent-width-matched composition the maximum sits at $5.42 R_p$ where $T = 1608$ K, against $T_{\text{max}} = 5043$ K at $1.45 R_p$: the temperature has fallen to 0.32 of its peak, which is the regime the mechanism asks for. The quench is visible too—in the advection-corrected profiles the density maximum has moved inward to $1.05\text{--}1.64 R_p$ and n_{2^3S} then falls monotonically, so the outflow appears to be sweeping the metastable out at least as fast as recombination resupplies it, and $\dot{m}_{2^3S}$ drops past the peak by 31% at the GJ 1132 equivalent-width-matched composition and by 98–99.9% on the helium-dominated $\text{He}/\text{H} = 1000$ runs, while the remaining cases carry it outward without loss.

Table 5: The metastable bump in the converged solutions. Columns 2–4 locate the n_{2^3S} maximum in the equilibrium profile and give the temperature and outflow speed there; the last three characterize the absorber-weighted ($w = n_{2^3S} r^2 dr$) distribution of $|v_r|$ over the advection-corrected 2^3S column that the transit tool integrates. The He/H = 1000 solutions have no outward density maximum—their n_{2^3S} peaks at the base—and are marked †. Those two rows are corrected values, recomputed on the advection-corrected profiles that the gate fix of Update_EXHALE.md Sect. 72 produced; the He/H ≤ 1 rows are unaffected by it.

case	r_{peak} [R_p]	T [K]	v [km s ⁻¹]	med. $ v_r $ [km s ⁻¹]	> 2 [%]	> 5 [%]
<i>GJ 1132 proxy SED</i>						
He/H = 0.55	5.42	1608	0.27	0.38	0.0	0.0
solar He/H	3.02	1876	0.07	0.41	0.0	0.0
He/H = 1000	1.05 [†]	2550	0.00	0.00	0.0	0.0
<i>GJ 699 proxy SED</i>						
He/H = 0.06	5.42	1015	1.40	2.53	64.9	0.0
solar He/H	5.50	1047	1.41	2.51	64.3	0.0
He/H = 1000	1.05 [†]	2529	0.00	0.00	0.4	0.0

The velocity condition, however, is not met, and not narrowly. At the bump the outflow is moving at 0.27 km s⁻¹ on the GJ 1132 proxy and 1.4 km s⁻¹ on the GJ 699 proxy. Weighting by absorbers over the whole 2^3S column, the median $|v_r|$ is at most 0.4 km s⁻¹ and 2.5 km s⁻¹ on the two spectra, and the largest $|v_r|$ anywhere in the column is 1.6 and 3.9 km s⁻¹. No part of the column on either spectrum reaches 5 km s⁻¹, against the $\sigma = 9.5$ km s⁻¹ of extra broadening the measurement demands (Sect. 8.2). Radial velocity is the generous choice here: the line-of-sight projection of a radial field over the transit chord is smaller still. Figure 9 shows the three statements together.

So the bump exists in these solutions and contributes nothing measurable to the width. That is consistent with what Schulik & Owen themselves report for the coolest hosts: for their M2 case they find that “the transit signal probes the same velocity region for both treatments of thermodynamics, making the line insensitive to velocity”, and they conclude that K stars, not M stars, are the favorable testbeds—G-type hosts suppress the bump by photoionizing the triplet, and M-type hosts have weaker tidal fields and so rarely drive fast enough outflows. LHS 1140 is an M4.5 dwarf, cooler than the M2 they model, and its planet is a temperate super-Earth rather than a hot giant. The mechanism appears to be present and correctly captured; the host is the wrong one for it to help.

8.6 This is a published shortfall

Taylor et al. (2026), with a model of the same class, reproduce the HD 189733 b line core but need ~ 12 km s⁻¹ of imposed non-thermal broadening for the width. Rumenskikh et al. (2022), in 3D, instead imposed reduced hydrogen cooling and 10–20 km s⁻¹ zonal jets. Schulik & Owen (2025) argue the width can come from a metastable population bump produced by adiabatic cooling, with no free parameter. The present result reproduces that pattern on a different,

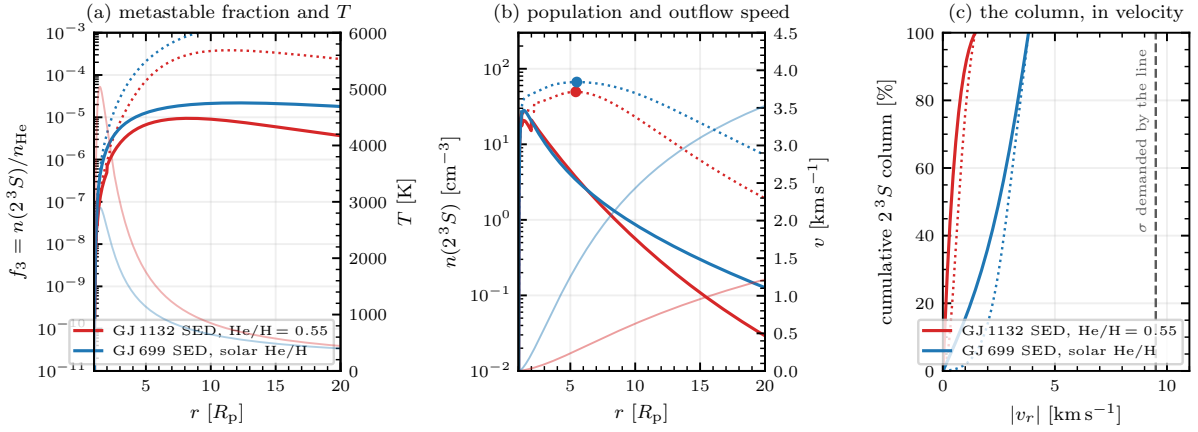


Figure 9: The adiabatic-cooling metastable bump in the EXHALE solutions (solid: advection-corrected, the profiles the transit integral uses; dotted: local equilibrium; faint curves on the right-hand axes). (a) The metastable fraction rises outward over the whole region where T falls, which is the bump in the sense of Schulik & Owen (2025). (b) The equilibrium n_{2^3S} peaks at 5.4 – $5.5 R_p$ (markers); advection moves the maximum inward and quenches the outer rise. (c) The absorber-weighted distribution of $|v_r|$ over the 2^3S column: the whole column lies below 4 km s^{-1} , far short of the $\sigma = 9.5 \text{ km s}^{-1}$ the measured width demands.

much cooler planet: a self-consistent energy balance produces a line far too narrow, and the literature’s responses are all ways of adding velocity that the 1D energy equation does not generate.

8.7 What would have to change

The measured line wants gas that is fast enough to be broad but not so extended that it absorbs too much. EXHALE at this XUV flux delivers neither. The candidates, in the order they should be tested:

1. *More energy* — tested (Sect. 10.), and it does not do it. The GJ 699 proxy carries five times the XUV of the GJ 1132 proxy used here, and the paper itself runs both. Solving the scan again on it raises $\dot{M}$ sevenfold and moves the composition estimate by an order of magnitude, but leaves the required extra broadening at 22.08 km s^{-1} against 22.30 : the width gap is not an energy-input problem.
2. *A different helium distribution* — tested (Sect. 8.3), and it does not do it either. Letting helium settle against binary diffusion and lifting it back with an eddy term redistributes the absorber by orders of magnitude and moves the inferred composition with it, but the required broadening stays at 22.31 km s^{-1} : it changes how much helium is there, not how fast it moves.
3. *Heating the code does not carry*. Taylor et al.’s trapped-ion turbulence, or any magnetically mediated dissipation, is outside a 1D hydrodynamic energy equation by construction.
4. *Geometry*. A line-of-sight velocity field from day–night or zonal flow is a 3D effect; Rumenskikh et al. reach the observed width with 10 – 20 km s^{-1} jets, which is the right order.

5. *Adiabatic-cooling population bump*. Looked for (Sect. 8.5): the bump is present in these solutions, but it sits at $0.3\text{--}1.4\text{ km s}^{-1}$, so it does not supply width.

The paper’s composition constraint should be read in the same light: it inherits the isothermal assumption. Impose a 6100 K structure and only helium-rich gas absorbs enough; solve the energy balance on this spectrum and, in the well-mixed limit, the same absorption is reached near $\text{He}/\text{H} \sim 0.5$ — and near 0.06 on the other proxy spectrum, or near 2 once helium is allowed to settle against the adopted eddy coefficient. No answer on that list is secure while the width is unexplained, because the width is what tells us whether the line-forming gas is the gas either model thinks it is.

9. The best each model can do

Figure 10 puts only the two best models on the measurement: the paper’s p-winds fit at its own best fit (the MCMC medians), and the EXHALE solution whose equivalent width matches. They fail in opposite directions. p-winds carries roughly the right width because its temperature is imposed where the line forms, but overshoots the blue component by a factor of about three. EXHALE carries the right total absorption but concentrates it into a line a third as wide as observed, so its core is too deep. Only the second failure can be undone by broadening alone (Sects. 8.2 and 8.4).

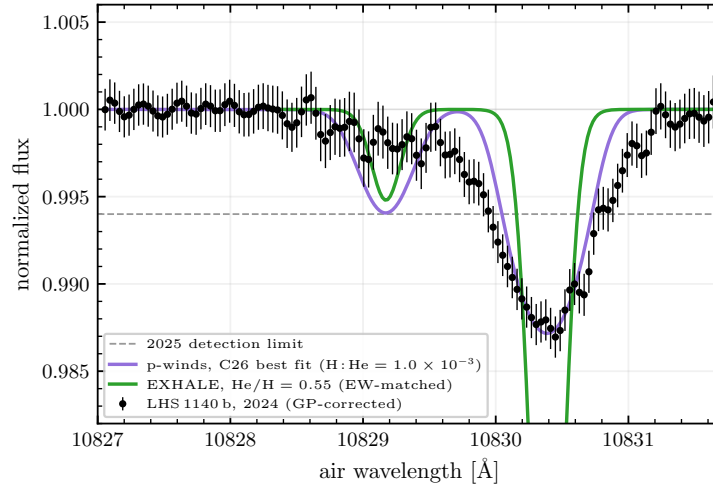


Figure 10: The two best models over the 2024 spectrum, in the presentation of the paper’s Fig. 4: p-winds at the C26 best fit (the MCMC medians) with the authors’ settings, and EXHALE at the equivalent-width-matched composition. Both are shifted by the fitted $v_{\text{wind}} = +2.26\text{ km s}^{-1}$.

Figure 11 shows what that EXHALE solution actually is. It is not a fast, hot outflow: the temperature peaks at 5043 K near $1.45 R_p$ — comparable to the isothermal value p-winds imposes — but then falls by adiabatic expansion to below 1000 K over the region that forms the line, and the velocity reaches only 1.2 km s^{-1} at $20 R_p$. Hydrogen is fully ionized above $\sim 3 R_p$ while helium stays largely neutral, and the metastable population peaks well out at $5.4 R_p$, $n(2^3S) \simeq 50\text{ cm}^{-3}$. That combination — a large, cold, slow, metastable-rich envelope — is precisely what produces the right

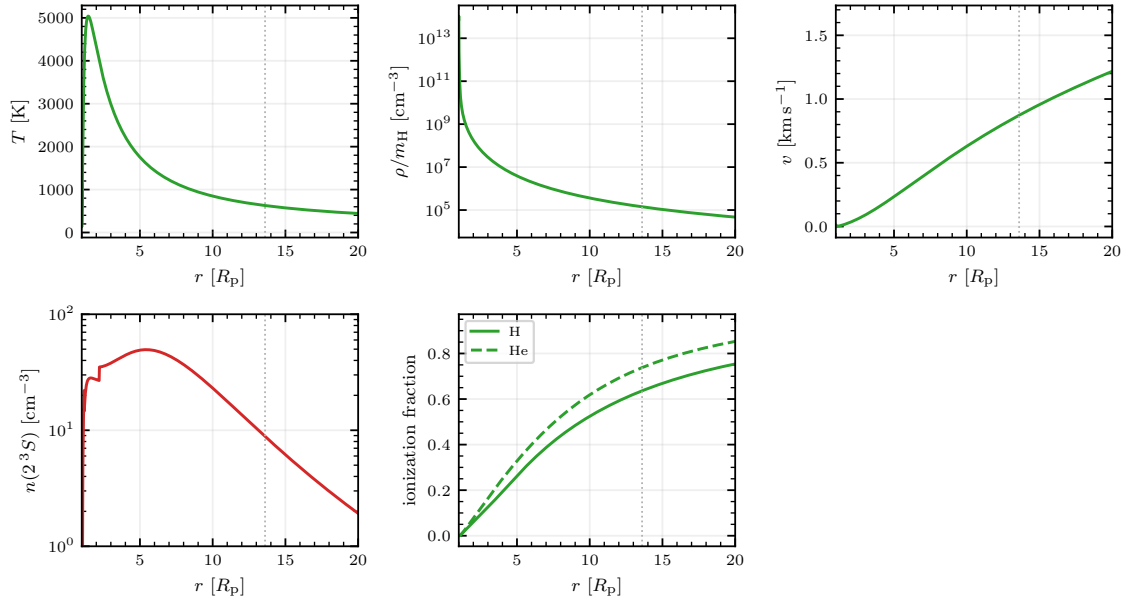


Figure 11: The EW-matched EXHALE solution ($\text{He}/\text{H} = 0.55$): temperature, density, velocity, metastable helium density, and ionization fractions. Dotted vertical line: the stellar radius.

equivalent width with the wrong profile: plenty of absorbers spread over most of the stellar disk, but almost no velocity spread to distribute the absorption in wavelength.

10. The other proxy: five times the XUV

The first item on the list above — more energy — can be tested without adding any physics, because the paper itself runs two proxy spectra. Both stand in for LHS 1140's unobserved XUV: GJ 1132 (used everywhere above) and GJ 699, scaled to the orbit by the catalog X-ray flux ratios $A_{1132} = 0.59$ and $A_{699} = 0.056$ following the authors' script, the same convention adopted in Sect. 2.3. Integrated over 10–1300 Å at the orbit of the planet, the GJ 699 file carries $169.9 \text{ erg s}^{-1} \text{ cm}^{-2}$ against 33.5 for the GJ 1132 one: a factor 5.1 in XUV input.

On the EXHALE side nothing else changes. Each GJ 699 case is the matching GJ 1132 input with one key replaced, `Spectrum file::`; the luminosity keys in the file are irrelevant because `sed_read` overwrites them with the integral of the spectrum it reads. Each case is started from the converged solution of the same composition on the GJ 1132 spectrum and finished with JFNK, and all seven reach `info = 0` with $\|R\| < 10^{-3}$. One case needed a different seed: $\text{He}/\text{H} = 0.55$ stalled when started from its GJ 1132 counterpart and converged when restarted from the already-solved $\text{He}/\text{H} = 0.25$ GJ 699 solution.

10.1 Escape rate: the retrieval value, on this spectrum

The brighter spectrum moves $\dot{M}$ where the composition never did:

	GJ 1132 proxy	GJ 699 proxy
$F_{\text{XUV}} (10\text{--}1300 \text{ \AA}) [\text{erg s}^{-1} \text{ cm}^{-2}]$	33.5	169.9
$\log_{10} \dot{M} [\text{g s}^{-1}], \text{He/H} \leq 1$	7.75–7.80	8.58–8.62
$\dot{M} [\text{g s}^{-1}]$	6×10^7	4.0×10^8
C26 retrieval on the same proxy	2.03×10^8	$4.22^{+1.14}_{-1.00} \times 10^8$

A factor 5.1 in XUV buys a factor $\simeq 7$ in $\dot{M}$ — steeper than linear, as expected when a larger fraction of the input goes into the wind rather than into radiation. The result is that on this spectrum EXHALE’s escape rate agrees with the paper’s retrieved value, $\log_{10} \dot{M} = 8.59$ against 8.63, inside 0.02–0.04 dex and far inside the retrieval’s own ± 0.11 dex. It is also 1.4 times the energy-limited estimate the paper quotes for this proxy, $2.90 \times 10^8 \text{ g s}^{-1}$. On the GJ 1132 spectrum the same code fell a factor 3 short of the retrieval. Whether a solved wind and an imposed one agree on $\dot{M}$ therefore depends on which proxy is assumed, and the disagreement reported in Sect. 8. is not a fixed property of the code.

10.2 Structure and line

Seven compositions were solved on this spectrum, two of them ($\text{He/H} = 0.04$ and 0.06) placed specifically to bracket the equivalent-width crossing:

He/H	$\log_{10} \dot{M}$ [g s ^{−1}]	T_{max} [K] (at r/R_p)	$v(20R_p)$ [km s ^{−1}]	$n(2^3S)_{\text{max}}$ [cm ^{−3}] (at r/R_p)	red [%]	FWHM [Å]	EW [% Å]
0.04	8.58	2605 (1.49)	3.61	30.9 (5.4)	2.65	0.270	0.761
0.06	8.59	2748 (1.48)	3.56	47.2 (5.4)	3.83	0.270	1.104
0.083	8.61	2896 (1.48)	3.53	66.7 (5.5)	5.21	0.272	1.507
0.25	8.62	3861 (1.44)	3.21	213 (5.9)	11.98	0.279	3.547
0.55	8.60	5079 (1.39)	2.83	441 (6.6)	16.64	0.288	5.092
1	8.59	6094 (1.38)	2.58	616 (7.2)	17.28	0.298	5.479
10 ³	8.43	9351 (1.38)	2.07	1.4×10^4 (1.0)	9.53	0.342	3.479
observed	—	—	—	—	1.254	0.841	1.108 ± 0.030

The wind is hotter and faster than on the fainter spectrum — at $\text{He/H} = 0.06$ it reaches 3.6 km s^{-1} at $20R_p$ against 1.2 for the equivalent-width-matched GJ 1132 solution — and the metastable population still peaks far out, near $5.4R_p$. As in Sect. 5., the $\text{He/H} = 10^3$ row is a corrected value (red $105.1\% \rightarrow 9.53\%$, EW $35.87 \rightarrow 3.479\%$ Å after the advection-gate fix, `Update_EXHALE.md` Sect. 72); the six helium-poorer rows, which are the ones the crossing is interpolated in, are unaffected.

10.3 The composition estimate moves by an order of magnitude

Interpolating the equivalent width in log–log across the seven solutions and solving for the measured value gives

$$\text{He/H} = 0.060 \quad (0.059\text{--}0.062 \text{ on the } EW \text{ } 1\sigma), \quad \text{H:He} = 16.6, \quad (6)$$

with the turbulence term on, 0.060 — again a shift below the percent level, so the crossing is set by the absorbing column and not by the broadening treatment. The bracketing runs are what fix this: the two compositions straddle the measured band directly ($EW = 0.761$ and 1.104 against $1.108 \pm 0.030 \text{ \% } \text{\AA}$), so the crossing is measured rather than extrapolated from the solar case.

This is a shift of one order of magnitude from the $\text{He}/\text{H} = 0.55$ the same code gives on the GJ 1132 proxy, and it crosses the solar value on the way: the brighter spectrum wants a helium-*poor* atmosphere, $0.72\times$ solar, where the fainter one wanted a helium-rich one, $6.6\times$ solar. The measurement cannot arbitrate, because the two proxies are two guesses at the same unobserved spectrum and both reproduce the observed equivalent width once composition is free. The conclusion to draw is therefore about the inference, not about the planet: what the helium line constrains is the product of composition and XUV input, and any composition quoted from it — the paper’s $\text{H}:\text{He} \lesssim 10^{-3}$ as much as the values here — is conditional on the assumed SED. On the present evidence the composition inference is SED-limited.

10.4 The width does not move

The width is the one thing the extra energy does not fix. Across the whole GJ 699 scan the red-pair FWHM is $0.270\text{--}0.342 \text{ \AA}$ ($0.304\text{--}0.427$ with the turbulence term), against the measured $0.841 \text{ \AA}$. Measuring the deficit the way Sect. 8.2 does — convolving with a Gaussian and asking which width reproduces the observation — the composition at the crossing needs 22.08 km s^{-1} FWHM of added velocity spread, against 22.30 km s^{-1} for the equivalent-width-matched GJ 1132 solution and 22.31 for the diffusive one of Sect. 8.3. A factor 5.1 in XUV input, a factor 7 in $\dot{M}$ and a factor 3 in outflow velocity at $20R_p$ change the required broadening by 1%.

The reason is visible in the table: the outflow velocity that grows is radial and small compared with the missing spread, while the line-forming region stays cold. The candidate “more energy” is therefore tested and rejected as an explanation of the width. It moves $\dot{M}$ and it moves the composition estimate; it does not move the profile. The remaining candidates — heating outside a 1D hydrodynamic energy equation, 3D velocity fields, and an adiabatic-cooling metastable bump — are unaffected by this test; the last of them has since been looked for on both spectra (Sect. 8.5) and does not help either.

11. Status against the goal

The line is not reproduced. What stands:

- **Amount of absorption: matched, under every assumption combination tried** — $\text{He}/\text{H} = 0.55$ well mixed on the GJ 1132 proxy, 0.060 well mixed on the GJ 699 proxy, 2.09 under diffusion at the adopted $K_{zz} = 10^9 \text{ cm}^2 \text{ s}^{-1}$ — and in each case insensitive to the broadening treatment at the 2% level. Which is the point of the next-to-last item: the match is always available, so it constrains the composition only as far as the assumptions behind it are secure.
- **Line shape: not matched**, by a factor 3.4 in both depth and width, and not fixable by composition. The missing ingredient is 22.30 km s^{-1} FWHM of velocity spread ($\sigma = 9.5 \text{ km s}^{-1}$), measured in Sect. 8.2 and Fig. 4; supplied as one Gaussian it puts depth, blue depth and red/blue ratio inside the observed 1σ interval at once, but its physical

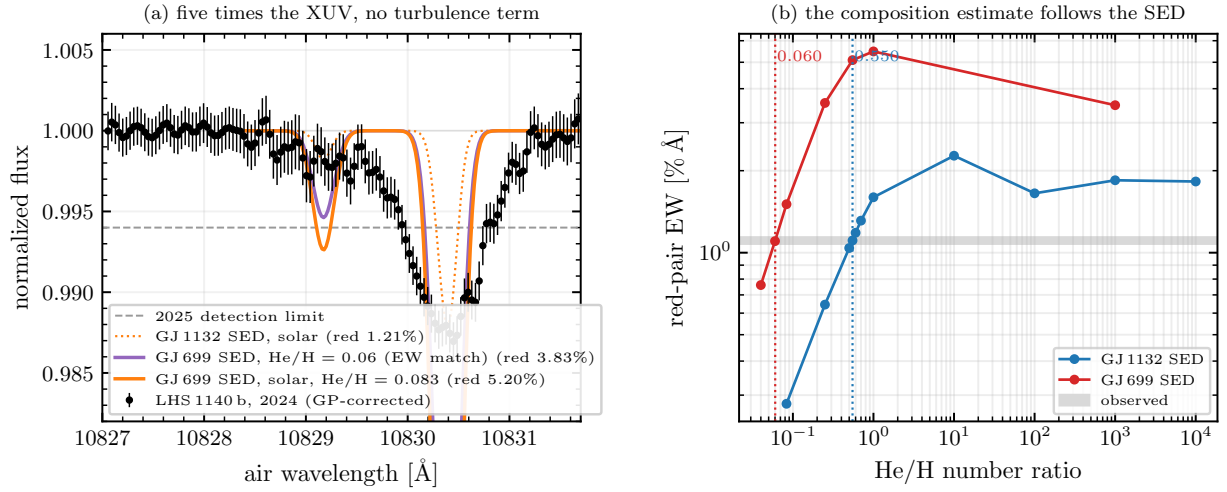


Figure 12: The GJ 699 proxy, five times the XUV of the GJ 1132 proxy used elsewhere in this memo. Left: the measurement with the EXHALE solution nearest the equivalent-width crossing ($\text{He}/\text{H} = 0.06$) and the solar case on this spectrum, with the GJ 1132 solar case dotted for contrast; all curves without the turbulence term and shifted by the fitted $v_{\text{wind}} = +2.26 \text{ km s}^{-1}$. The lines are deeper and no wider. Right: red-pair equivalent width against composition for the two spectra with the measured band and the two crossings; the composition the line implies moves from 0.550 to 0.060 when the assumed SED is changed.

origin is unidentified. The demand is unchanged under diffusion and the adopted K_{zz} : measured the same way on the EW-matched diffusive solution it is 22.31 km s^{-1} , 0.01 km s^{-1} away (Sect. 8.3), and it is 22.08 km s^{-1} on the $5.1 \times$ brighter proxy spectrum (Sect. 10.). The adiabatic metastable bump, which would have supplied it without a free parameter, is present and measured (Sect. 8.5) but sits too slow to contribute.

- **Escape rate: an output, not a fit, and SED-set.** $\dot{M} \simeq 6 \times 10^7 \text{ g s}^{-1}$ on the GJ 1132 proxy, a third of the value the retrieval imposes, and nearly independent of composition. On the GJ 699 proxy, $5.1 \times$ brighter in XUV, it is $4.0 \times 10^8 \text{ g s}^{-1}$ — a factor 7 higher and within 0.04 dex of the value the paper retrieves on that same proxy (Sect. 10.).
- **Composition: SED-limited, and weakly K_{zz} -limited.** In the well-mixed limit the equivalent-width crossing is $\text{He}/\text{H} = 0.550$ on one proxy spectrum and 0.060 on the other, helium-rich against helium-poor, with the measurement unable to separate them. With binary H/He diffusion active the crossing rises, and the whole K_{zz} axis has now been walked: $\text{He}/\text{H} = 4.68$ on the plateau at $K_{zz} \leq 10^5$, then 4.45, 3.78, 2.87, 2.09, 1.54 and 1.11 at 10^6 through $10^{11} \text{ cm}^2 \text{ s}^{-1}$ (Table 2, Fig. 5) — a factor 1.9 across the range the literature leaves open, 4.2 across the entire axis, against the factor 9 the SED carries. End to end the inference spans 0.060 to 4.68, a factor 80. So any composition read off this line, the paper’s included, is strongly conditional on the assumed XUV spectrum and weakly conditional on the eddy coefficient, as $K_{zz}^{-0.14}$ wherever the eddy term acts at all.
- **One solution is not converged to tolerance.** $\text{He}/\text{H} = 100$ stagnates at $\|R\| = 1.18 \times 10^{-3}$ from every seed tried; neighbours bracket it smoothly, so it is retained with the residual recorded.

Three axes have now been walked — the assumed XUV spectrum, the mixing assumption with its eddy coefficient, and composition itself — and each separates the two questions rather than answering them together. Every one of them moves the composition the equivalent width implies, two of them by a large factor; none of them moves the width. What is left as a possible source of the missing 22 km s^{-1} is therefore what it was: velocity fields that are three-dimensional, or heating a 1D hydrodynamic energy equation does not carry. The adiabatic-cooling metastable bump, the one candidate that would have needed no free parameter, has been looked for on both spectra and does not supply it for this host (Sect. 8.5).